

Name: _____



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 2

Class Reg Number

17

Calculator Model / No.

Chemistry

9729/02

Paper 2 Structured Questions

18 September 2018

2 hours

Additional Materials: *Data Booklet*

INSTRUCTIONS TO CANDIDATES

Write your name, class and register number in the spaces provided at the top of this page.

Write your calculator brand and model/number in the box provided above.

Answer **all** questions in the spaces provided on the question paper.

All working must be shown clearly.

INFORMATION FOR CANDIDATES

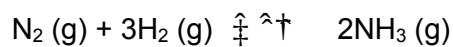
The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

Examiner's Use			
Paper 1	MCQ	/ 30	/ 15 %
Paper 2	Q1	/ 13	
	Q2	/ 19	
	Q3	/ 14	
	Q4	/ 15	
	Q5	/14	
	Total	/ 75	/ 30%
Paper 3	Total	/ 80	/ 35%
Paper 4	Total	/ 55	/ 20%
Total	/ 100 %		
Grade			

This document consists of **17** printed pages (including this cover page).

- 1 Ammonia is one of the most highly produced inorganic chemicals. Modern chemical plants depend on Haber Process whereby hydrogen derived from natural gas reacts with atmospheric nitrogen using a catalyst under high pressure and temperature to produce anhydrous ammonia.



- (a) For the above reaction, a 1:3 ratio of N_2 and H_2 is passed over a catalyst. The resulting mixture is allowed to reach dynamic equilibrium. At equilibrium, 20% of the N_2 has reacted and the total pressure is 2.0 atm.

- (i) Explain what is meant by the term *dynamic equilibrium*.

[1]

- (ii) Calculate the partial pressures of N_2 and H_2 at equilibrium.

[2]

- (iii) Write an expression for the equilibrium constant, K_p , and calculate its value.

[2]

- (b) (i) Using relevant data in the *Data Booklet*, calculate the enthalpy change of reaction, ΔH_{rxn} of the Haber Process.

[2]

- (ii) The enthalpy change of reaction calculated above in (b)(i) differs slightly from the actual value of $-92.0 \text{ kJ mol}^{-1}$. Suggest a reason for this.

[1]

- (c) Explain whether the formation of ammonia is favoured with

- I high or low pressure,
II high or low temperature.

[3]

- (d) Ammonium chloride, NH_4Cl is typically used in instant ice-packs. A crushing action activates the ice-pack as this mixes NH_4Cl with water. The ice-pack cools as the dissolution of NH_4Cl is endothermic.

Draw an energy level diagram to determine the standard enthalpy change of solution of ammonium chloride given the following information.

Lattice energy of ammonium chloride	-705 kJ mol^{-1}
Enthalpy change of hydration of NH_4^+	-307 kJ mol^{-1}
Enthalpy change of hydration of Cl^-	-381 kJ mol^{-1}

[2]

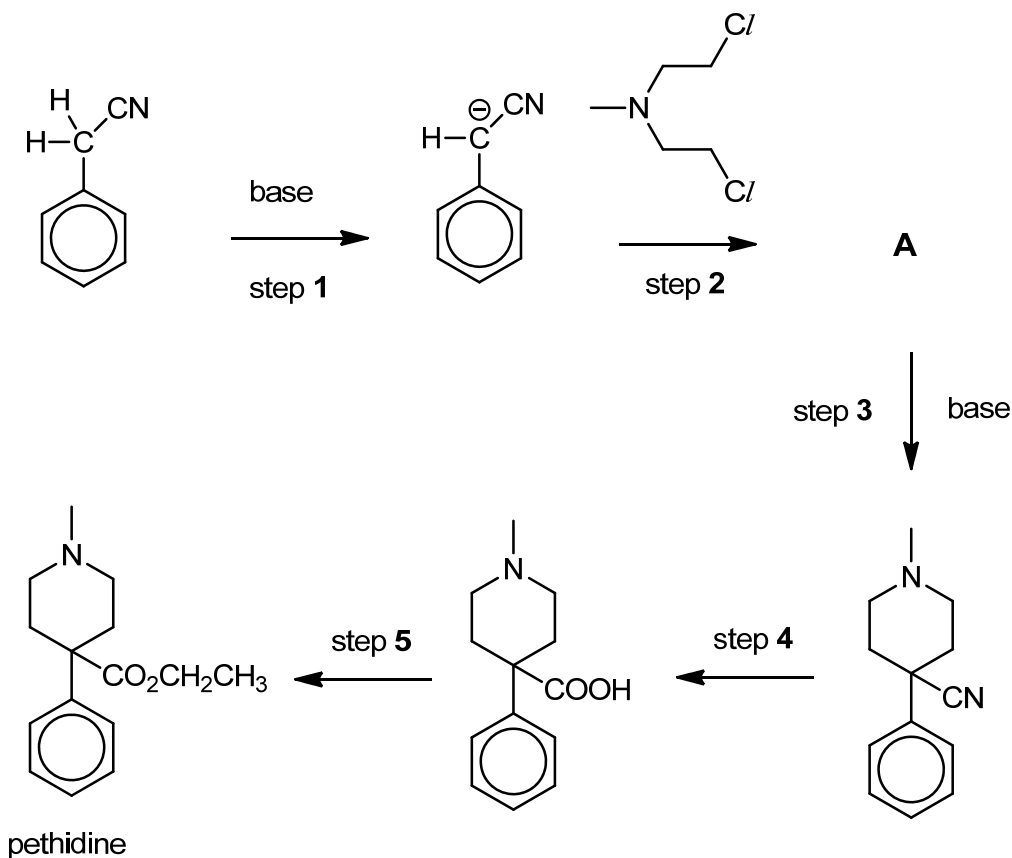
Energy



[Total: 13]

2 Nitrogen is an element commonly found in various major classes of drugs.

(a) The following is a reaction scheme for the synthesis of pethidine, a powerful painkilling drug.



(i) State the type of reaction for step 1 and step 2.

[2]

Step 1:

Step 2:

(ii) Suggest the structure of intermediate **A**

[1]

(iii) State the reagents and conditions needed for step 4 and step 5.

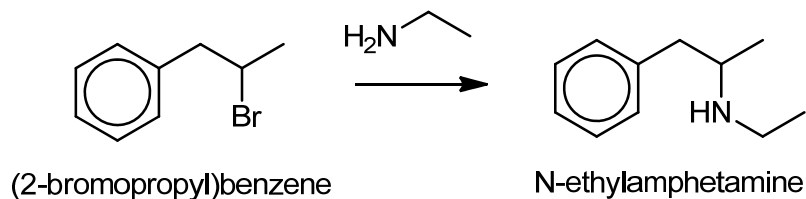
[2]

Step 4:

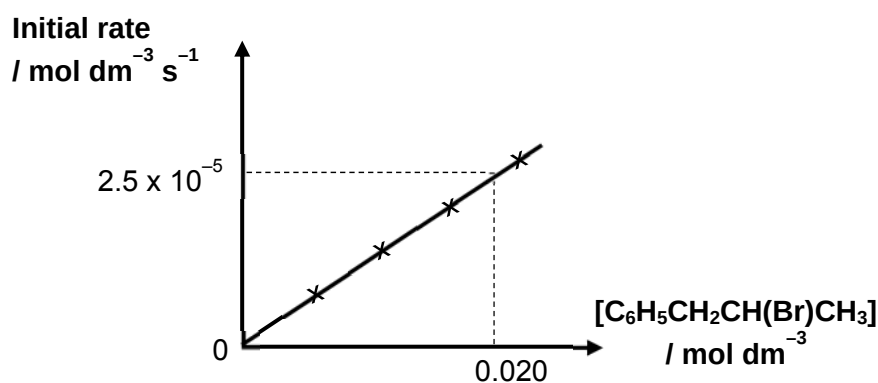
Step 5:

- (b) Kinetics studies of organic reactions provide strong evidence in supporting the postulated mechanisms.

The results from the kinetics study of the nucleophilic substitution reaction involved in the synthesis of N-ethylamphetamine, a discontinued weight-loss drug, are presented below.



The graph was plotted based on data obtained from repeating the experiment several times by varying the concentration of $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{Br})\text{CH}_3$ while maintaining that of $\text{CH}_3\text{CH}_2\text{NH}_2$ to be at 2.5 mol dm^{-3} .



- (i) The rate equation for the reaction was found to be

$$\text{rate} = k [\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{Br})\text{CH}_3][\text{CH}_3\text{CH}_2\text{NH}_2]$$

Using the graph provided, justify why the order of reaction with respect to $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{Br})\text{CH}_3$ is one.

[2]

- (ii) Calculate the rate constant, k and state its units.

[2]

- (iii) Given that this is a bimolecular reaction, outline the mechanism for this reaction. In your answer, show any relevant charges, lone pairs of electrons and movement of electrons.

[3]

- (iv) Suggest why the rate of reaction decreases when $(\text{CH}_3)_3\text{CNH}_2$ is used instead of $\text{CH}_3\text{CH}_2\text{NH}_2$.

[1]

- (v) Explain how the reactivity of (2-chloropropyl)benzene will differ towards $\text{CH}_3\text{CH}_2\text{NH}_2$ as compared to (2-bromopropyl)benzene.

[1]

- (vi) With the aid of a Maxwell–Boltzmann Distribution curve, explain how heating the reaction mixture will increase the rate of the reaction.

[3]

- (c) Investigating the optical activity of the product of a nucleophilic substitution reaction can also provide evidence on how the reaction may proceed.

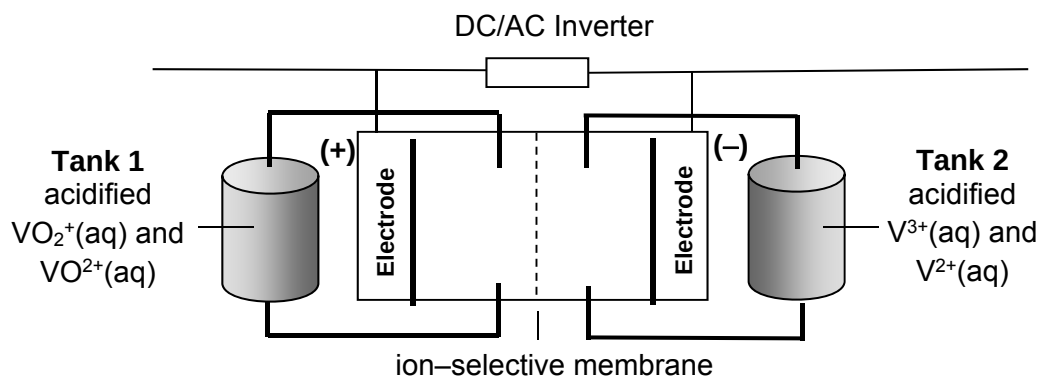
Consider the reaction between the (+) enantiomer of 2–bromobutane and hot NaOH (aq).

State the expected optical activity of the product if this reaction proceeds via S_N1 mechanism. Provide reasoning for your answer.

[2]

[Total: 19]

- 3 The vanadium flow battery (VFB) is a type of rechargeable battery that employs electrolyte containing vanadium ions in different oxidation states.



- (a) Electricity is generated from the battery by half-cell reactions of these electrolytes that were pumped into the cell from separate storage tanks. The two electrolytes in the cell are separated by an ion selective membrane, which only allows H^+ ions to pass through.

- (i) Write the two half equations for the reaction occurring at the anode and the cathode when the battery discharges. Hence, write the overall cell reaction.

[2]

Anode:

Cathode:

Overall:

- (ii) Calculate the e.m.f of the battery.

[1]

- (iii) Explain how the ion-selective membrane in VFB maintains electrical neutrality.

[1]

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- (iv) State and explain what happens to the overall e.m.f when a small amount of OH^- (aq) ions were added to the electrolyte in **Tank 1**.

[2]

- (b) The total concentration of vanadium ions in the 20 dm^3 electrolyte in **Tank 1** is 5.00 mol dm^{-3} . During the charging of the VFB, the percentage composition of VO_2^+ has to be increased from 5% to 90%.

- (i) Calculate the amount of VO_2^+ that needs to be converted during this charging process.

[1]

- (ii) Find the time taken, in hours, if a 120 A current is used in the charging process.

[2]

(c) Anodising of aluminum is an electrolytic process to increase the thickness of the aluminium oxide layer on the surface of aluminium, by passing a direct current through dilute sulfuric acid.

(i) Draw a labelled diagram of an electrolysis set-up during the anodising of aluminium.

[2]

(ii) Write the half-equations for the reactions taking place at the anode, cathode and hence write the overall equation.

[2]

Anode:

Cathode:

Overall:

(iii) State and explain any observations if copper(II) sulfate was used as the electrolyte instead.

[1]

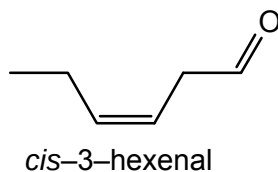
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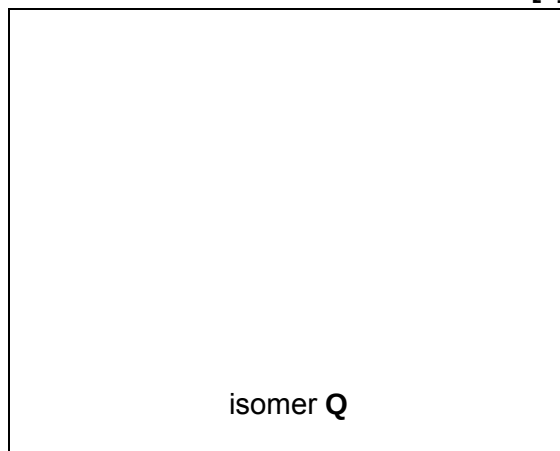
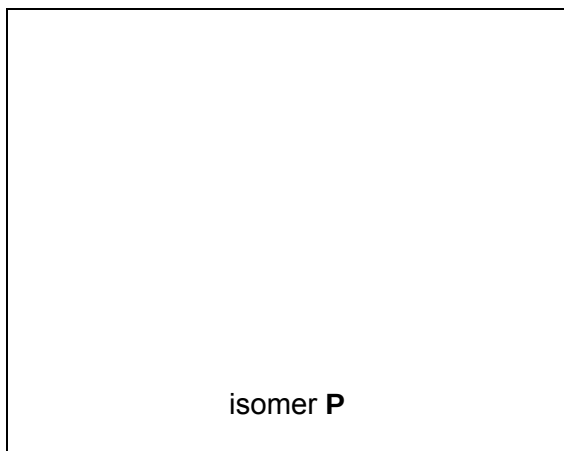
- 4(a) *cis*-3-hexenal, $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CHO}$ is a compound with an intense smell of freshly cut grass and leaves.



Two isomers, **P** and **Q**, with the same molecular formula, $\text{C}_6\text{H}_{10}\text{O}$, as *cis*-3-hexenal show different types of isomerism. Both isomers decolourises aqueous bromine in the dark and both gives a positive test with Fehling's reagent.

Draw their structures and state their isomeric relationship to *cis*-3-hexenal.

[2]



isomeric relationship to *cis*-3-hexenal:

isomeric relationship to *cis*-3-hexenal:

- (b) Aldehydes and ketones (broadly classified as carbonyl compounds) can be reduced to alcohols using metal hydride reagents such as lithium aluminium hydride (LiAlH_4) and sodium borohydride (NaBH_4). These metal hydride reagents provide a source of hydride, H^- ions.
- (i) The reaction between *cis*-3-hexenal, $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CHO}$ and sodium borohydride in methanol is a nucleophilic addition reaction.

Draw the mechanism for this reaction, showing all charges, dipoles and curly arrows to represent the movement of electron pairs.

[3]

- (ii) BH_3 is a side product of the earlier reaction in (b)(i). BH_3 can form an addition product with Lewis bases, for example, with the alkoxide ion generated in the mechanism.

Draw the structure of this adduct formed and explain why the formation of the adduct is likely to happen.

[2]

- (c) Both LiAlH_4 and NaBH_4 are sources of H^- . Hydride ions react vigorously with Bronsted–Lowry acids, releasing H_2 (g) and heat.

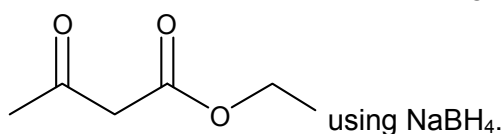
LiAlH_4 reacts violently with water or alcohols. When carrying out reactions with LiAlH_4 , chemists often use solvents like anhydrous diethyl ether that has been treated to remove traces of water. On the other hand, NaBH_4 does not react that vigorously with water or alcohol.

Using the information provided or otherwise, suggest why LiAlH_4 is a better source of H^- than NaBH_4 .

	Pauling Electronegativity
aluminium	1.61
boron	2.04
hydrogen	2.20

[1]

- (d) A student proposed the following procedure to reduce ethyl acetoacetate ($\text{C}_6\text{H}_{10}\text{O}_3$),



Procedures

1. Add 0.70 g of NaBH_4 to 25 cm^3 ethanol solvent in a 100 cm^3 round-bottomed flask. Cool the resulting mixture to 0 °C using an ice-bath.
2. To this mixture, slowly add 30 cm^3 of 40.8 g dm^{-3} ethyl acetoacetate solution. Stir the resulting solution at 0 °C for 15 minutes, then allow to warm to room temperature and stir for an additional 15 minutes.
3. Evaporate the solvent. Dissolve the resulting white solid in 30 cm^3 dichloromethane solvent, CH_2Cl_2 .
4. Cool the flask in an ice-bath. Then, add 30 cm^3 of 1 mol dm^{-3} hydrochloric acid dropwise, while stirring, to quench the reaction by destroying any unreacted hydride.
Safety note: The addition of HCl will release H_2 gas.
5. Separate the organic layer from the aqueous layer.
6. Add powdered magnesium sulfate to the organic layer in excess, till no clumping of magnesium sulfate powder is observed. Filter off the magnesium sulfate and evaporate the dichloromethane solvent.
7. Weigh the product and record the actual yield.

- (i) Explain why NaBH_4 is soluble in ethanol (in **step 1**) in terms of structure and bonding.

[2]

- (ii) One mole of NaBH_4 can reduce a maximum of four moles of carbonyl groups. Generally, in an experimental procedure, the reducing agent NaBH_4 should be used in excess.

Determine with relevant calculations, whether the student is correct with the quantities proposed in his procedures (**step 1 – 2**).

[2]

- (iii) In **step 4**, hydrochloric acid is added to quench the reaction, producing hydrogen gas and sodium boron chloride. Write an equation for the reaction that took place.

[1]

- (iv) Draw the **skeletal** formula of the product isolated in the organic layer at the end of the experiment.

[1]

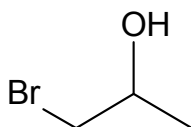
- (v) Suggest the function of magnesium sulfate in **step 6**.

[1]

[Total: 15]

- 5 Organic reactions can lead to increased variation due to the direction-specific nature of covalent bonds, giving rise to much complexity from using only a small range of elements such as carbon, hydrogen and oxygen.

- (a) Propene reacts with aqueous bromine to form a mixture of products, including molecule **A**.



Molecule **A**

- (i) State the IUPAC name of molecule **A**.

[1]

-
- (ii) Draw a mechanism for the reaction between propene and aqueous bromine to form Molecule **A** and hydrogen bromide. Show relevant lone pairs and dipoles, and use curly arrows to indicate the movement of electron pairs.

[3]

- (iii) Molecule **A** is the major product of this reaction. Suggest the structure of two other organic by-products.

[2]

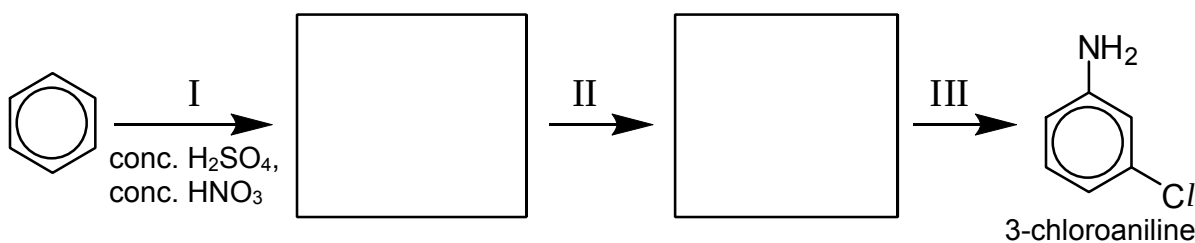
- (iv) In the mechanism outlined in (a)(ii), two stereoisomers of **A** are formed in equal amounts. Draw these two stereoisomers and name the type of isomerism displayed.

[2]

- (b) 3-chloroaniline can be synthesised from benzene in a 3-step synthesis.

- (i) Propose a synthetic route by suggesting the two intermediate compounds and state the reagents and conditions used for steps II and III.

[4]



Step II:

Step III:

- (ii) One of the by-products from the synthesis in (b)(i) is HCl, which is a strong acid. The acid dissociation constant values, K_a of several hydrogen halides are shown below.

Compound	$K_a / \text{mol dm}^{-3}$
HCl	1.3×10^6
HBr	1.0×10^9
HI	3.2×10^9

Account for the relative K_a values.

[2]

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[Total: 14]

END OF PAPER